

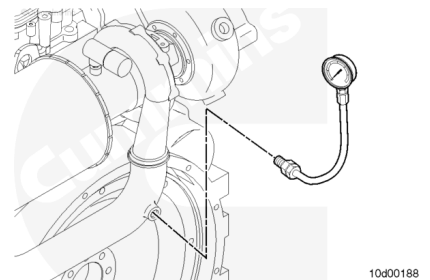
## Trainings ▾

### Initial Check

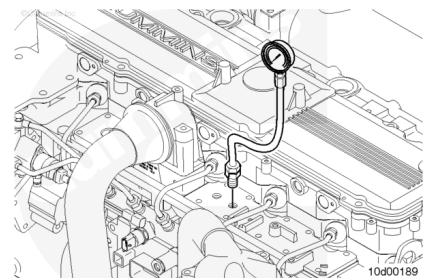
#### Test

**Note : For marine seawater aftercooler only.**

Install pressure gauge, Part Number ST-1273, in the turbocharger boost pressure fitting in the elbow at the turbocharger outlet.

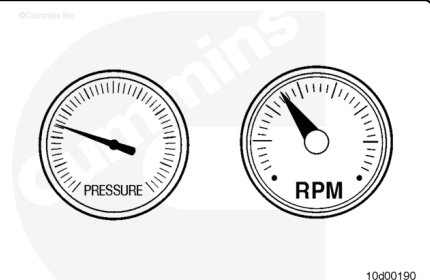


Install another pressure gauge, Part Number ST- 1273, in the intake manifold on the outlet side of the aftercooler.



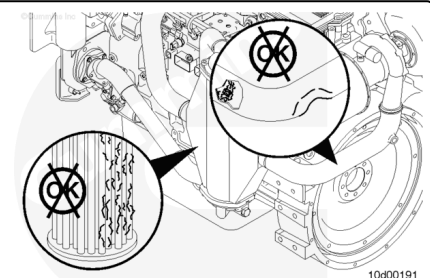
Operate the engine at rated rpm and load. Record the readings on the two gauges.

| Aftercooler - Pressure Differential |          |
|-------------------------------------|----------|
| 380 mm Hg                           | 15 in Hg |



If the pressure differential is greater than the specification, inspect the aftercooler and associated piping for plugging.

Clean or replace if necessary.

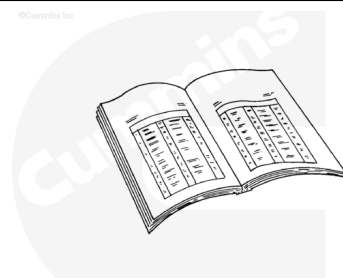


# Preparatory Steps

## Automotive and Industrial

### **⚠ WARNING ⚠**

**Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.**



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### **⚠ WARNING ⚠**

**Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.**

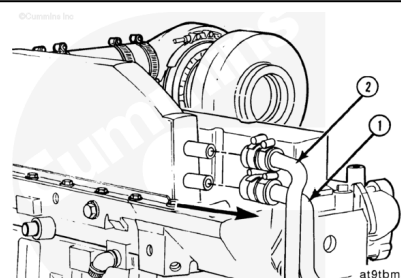
Disconnect the cold starting aid, if used. Refer to the OEM service manual.

Remove the air crossover tube. Refer to Procedure 010-019 (</qs3/pubsys2/xml/en/procedures/100/100-010-019.html>).

Drain 2 liters [2 qt] of coolant. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/100/100-008-018.html>).

## Remove

Remove the coolant supply tube (1) and the coolant return tube (2) from the aftercooler.

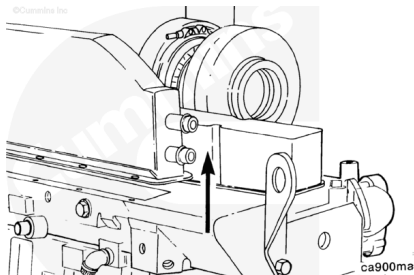


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Remove the aftercooler housing mounting capscrews.

Remove the aftercooler housing and gasket.

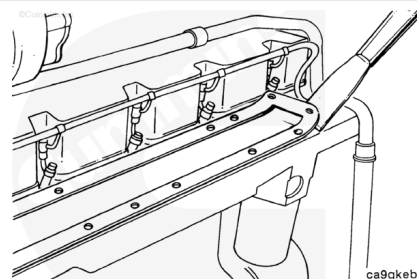
**Note :** For a form-in-place gasket, cut the aftercooler gasket to remove.



## Clean

Clean the aftercooler housing sealing surfaces.

Keep the gasket material and all other materials out of the intake manifold.



## Install

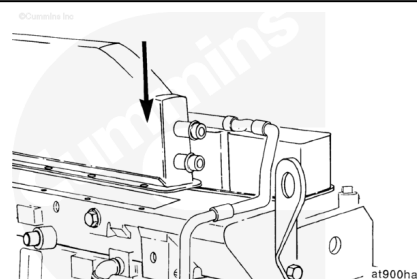
Use silicone sealant, Part Number 3823494, on both sides of the aftercooler housing gasket.

Install the aftercooler housing and new gasket.

**Note :** For aftercoolers with a form-in-place gasket, apply a 4-mm bead of sealant to the cylinder head.

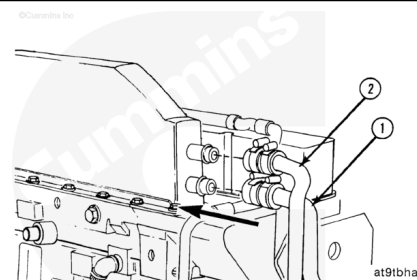
Install and tighten the aftercooler housing mounting cap screws.

**Torque Value:** 24 n•m [ 18 ft-lb ]



Install the coolant supply tube (1) and coolant return tube (2).

**Torque Value:** 5 n•m [ 44 in-lb ]



## Finishing Steps

### **⚠ CAUTION ⚠**

**During filling, air must be vented from the engine coolant passages. Open the engine vent petcock. Make sure to open the petcock on the aftercooler for aftercooled engines. The system must be filled slowly to prevent air locks. Wait two to three minutes to allow air to be vented; then add coolant to bring the level to the bottom of the radiator filler neck.**



Connect the cold starting aid, if used. Refer to the OEM service manual.

Install the air crossover tube. Refer to Procedure 010-019 (</qs3/pubsys2/xml/en/procedures/100/100-010-019.html>).

Fill the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/100/100-008-018.html>).